

Correction Responsiveness versus Complete-Trajectory Performance in Interactive PET/CT Lesion Segmentation

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Abstract. The autoPET V challenge studies iterative lesion segmentation in whole-body PET/CT from foreground and background scribbles. We developed a transfer-based system that starts from a strong LesionLocator segmentation and uses a frozen autoPET IV interactive model only as a correction proposer. For each cumulative prompt update, the model is evaluated before and after the new scribble; only the target-direction binary difference is composed onto an externally maintained mask. On a 24-case development cohort, this state-preserving formulation improved mean Dice from 0.805964 at the initializer to 0.827897 after five corrections and mean Detection-Matching Metric (DMM) from 0.807865 to 0.842000, while preserving prior corrections. We separately investigated V-native scratch training. Scaling from 48 patients to 1,477 studies enabled lesion segmentation and bidirectional scribble conditioning, and a second-generation self-error curriculum increased selected hard-residual rescue. However, these local response gains did not improve complete official-style trajectories. On the development cohort, adding the learned corrector reduced both primary area-under-trajectory metrics. In a single prespecified comparison on a patient-disjoint 24-case confirmation cohort, adding a second frozen corrector produced nearly unchanged AUC-Dice (+0.000771) but materially lower AUC-DMM (-0.177125). We therefore selected the single frozen corrector. The results show that correction responsiveness and complementarity are insufficient selection endpoints: interactive systems should be evaluated over complete, stateful trajectories with lesion-level as well as voxel-level metrics.

Keywords: PET/CT · interactive segmentation · scribble prompts · lesion segmentation · state preservation

Table 1. Algorithm details in the format required by the autoPET V publication template.

Team name	algorithm name (as submitted on grand-challenge)	data pre-processing	data post-processing	training data augmentation
shoheyhey53	autoPETV final	Geometry check; released nnU-Net normaliza- tion/resampling; native restoration; multipoint radius-2 EDT prompts	threshold 0.5; target-direction binary prompt delta composed onto the current mask	N/A – no new training or fine-tuning
test time augmentation	ensembling (e.g. cross-validation, model ensemble, ...)	standardized framework? (e.g. nnUNet, MONAI, ...)	network architecture (e.g. UNet (3D))	loss
None; mirroring disabled	Step0: mean model-space logits from released folders [0, 5]; corrector: single released folder 0	nnU-Net v2.6.1; PyTorch 2.7.1	3D LesionLocator plus 3D residual-encoder U-Net corrector	N/A – no training or fine-tuning of the submitted system
training data	data/model dimensionality and size (e.g. 2D: 128x128, 3D: 128x192x160, ...)	use of pre-trained models (public available or own developed)	GPU hardware for training	
No new training; public pretrained models were used	3D volumes; released 3D full-resolution plan uses a $192 \times 192 \times 192$ patch; native-size output	Yes; public autoPET- interactive source and released model folders, referenced upstream with exact hashes	N/A for submitted system	

1 Introduction

Whole-body PET/CT lesion segmentation is complicated by heterogeneous lesion appearance, physiological tracer uptake, and the differing distributions of fluorodeoxyglucose (FDG) and prostate-specific membrane antigen (PSMA) examinations. Automatic segmentation can provide a useful starting point, but an interactive system must also translate corrective input into reliable state changes without destroying already correct regions. The autoPET V task formalizes this setting through an initial prediction followed by corrective foreground (FG) and background (BG) scribbles, and gives equal importance to voxel overlap and lesion-level detection matching [1].

This setting creates a distinction that is easy to overlook. A promptable network may produce a plausible complete mask, yet replacing the current state with that mask can undo earlier corrections or introduce unrelated changes. Conversely, a model may respond correctly to selected local errors without improving a complete multi-interaction trajectory. Our central hypothesis was that a strong pretrained interactive PET/CT model could be transferred while preserving the current state by separating *proposal* from *state update*. We therefore maintain the current mask outside the network, extract only the prompt-induced target-direction binary response, and compose that response onto the existing state.

We make five restrained contributions. First, we evaluate zero-shot transfer of a pretrained interactive PET/CT corrector to autoPET V. Second, we formulate a state-preserving incremental prompt-delta composer. Third, we evaluate complete M_0 – M_5 trajectories with the official Dice and DMM implementations. Fourth, we investigate V-native three-phase and second-generation self-error curricula. Fifth, we show empirically that improved local correction response and model complementarity need not translate into improved complete trajectories.

2 Related Work

Our initializer and corrector build on the nnU-Net family [2, 5]. Rokuss et al. extended a multitracer PET/CT segmentation pipeline with FG/BG point channels and online interaction simulation; local Euclidean-distance-transform (EDT) prompt encodings were particularly effective [4]. We reuse their public interactive checkpoint without fine-tuning.

Two directly relevant autoPET V reports appeared during the final development period. Zhang describes a large residual-encoder U-Net trained with automatic, ground-truth-scribble, and error-driven phases over autoPET and DeepPSMA data, followed by five-fold checkpoint ensembling [6]. Lozano-Jimenez et al. describe anatomy-aware supervision, online interactive training, and tracer-dependent routing [3]. Our V-native experiments use a related staged principle but are single-fold, time-boxed studies on official autoPET data only. The submitted system differs more fundamentally: it does not fine-tune the corrector and applies only its incremental binary prompt response to an external mask.

Table 2. Cohorts and roles. Development counts are cases unless stated otherwise.

Cohort	Role	FDG PSMA Lesion status		
Development cohort (24)	mechanism development and selection	12	12	17 positive, 7 negative
Patient-disjoint confirmation cohort (24)	single prespecified baseline-versus-challenger comparison	12	12	16 positive, 8 negative
Small scratch	train/dev patients (48/16)	mixed mixed mixed		
Expanded scratch	train studies/protected dev patients (1,477/64)	955	522	train: 947 positive, 530 negative

3 Materials and Evaluation

3.1 Cohorts and leakage control

Table 2 summarizes the cohorts. A 24-case development cohort supported mechanism development and selection. A patient-disjoint 24-case confirmation cohort was accessed only for one prespecified baseline-versus-challenger comparison after the initializer, challenger, threshold, members, and fusion rule had been frozen. It was not used for threshold, member, or fusion tuning. The expanded V-native experiment used 1,477 official studies from 1,166 patients; DeepPSMA was not used. Patient identifiers are neither reported nor included in the paper artifacts.

3.2 Interactions and metrics

The canonical local trajectory contains states M_0 – M_5 . At each correction, a current-error FG or BG scribble is generated, converted to at most ten source points, encoded with the public radius-2 local EDT semantics, and accumulated with prior prompts. Dice measures voxel overlap. For DMM, lesions are 18-connected components; a ground-truth component is detected when any predicted component reaches intersection-over-union at least 0.1, multi-assignment is not penalized, and the resulting TP/FP/FN counts form an F1 score. False-negative volume (FNV) and lesion-negative false-positive volume (FPV) are reported in milliliters. Raw trajectory AUC is the trapezoidal integral across all six states.

All official-style metric calculations use the challenge repository pinned at commit [4a2026866bfacc812492cfc7e6a8c54ac3c4f703](#) [1]. Exact hidden-server step depth and leaderboard aggregation were not available locally; consequently, every AUC claim in this paper denotes the pinned local six-state implementation, not a hidden leaderboard score.

Table 3. Frozen final system configuration.

Component	Frozen configuration
Initializer	LesionLocator released folders [0,5]; model-space logit average; threshold 0.5
Corrector	autoPET IV <code>ResidualEncoderUNetOrgan</code> , released folder 0; CT/PET/FG/BG inputs
Prompt encoding	cumulative multipoint-10 conversion; public local radius-2 EDT; voxelwise union
State	externally maintained binary mask; no previous-mask network channel
Update	only the new target-direction binary prompt delta is composed
Learning	none: no optimizer, backward pass, adapter, or new checkpoint

4 Methods

4.1 Data pre-processing and geometry

The container accepts one CT MHA volume, one PET MHA volume, and cumulative lesion-click JSON. CT/PET size, spacing, origin, and direction must match and all arrays must be finite. SimpleITK arrays are transformed to the released nnU-Net convention; the released plans control resampling and intensity normalization. Predictions are restored to the native MHA geometry. New scribbles are ordered deterministically on their native slice, sampled by voxel-space arc length to at most ten distinct points, accumulated by polarity, and converted by the public corrector to radius-2 EDT channels. No data-dependent post-processing sweep is used.

4.2 Final submitted system

The final system (Table 3) uses a two-checkpoint LesionLocator initializer ensemble, one frozen interactive corrector, and an external state transition. CT and PET first enter the released LesionLocator folders 0 and 5. Their model-space logits are averaged and restored to native geometry to form M_0 at threshold 0.5. The corrector is the public autoPET IV `ResidualEncoderUNetOrgan` released-folder-0 model. Its exact checkpoint SHA-256 is

`a0cb3a89c72b0a79a27900980361385ff02572c0c71aba6609390fecbbc13e82`.

It receives CT, PET, an FG prompt channel, and a BG prompt channel, and predicts a whole lesion mask. Its weights remain immutable.

4.3 State-preserving prompt-delta composition

Let X denote CT/PET, M_{t-1} the current external mask, and P_{t-1} and P_t the cumulative prompt states immediately before and after a new scribble. With

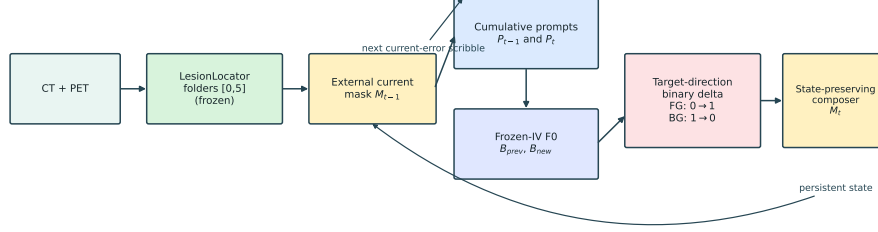


Fig. 1. Final inference pipeline. Frozen networks produce an initializer and paired prompt-conditioned proposals; the target-direction binary difference is composed onto persistent external state.

frozen corrector f and threshold $\tau = 0.5$,

$$B_{\text{prev}} = \mathbf{1}[f(X, P_{t-1}) \geq \tau], \quad B_{\text{new}} = \mathbf{1}[f(X, P_t) \geq \tau]. \quad (1)$$

For an FG correction,

$$\Delta_{\text{FG}} = \neg B_{\text{prev}} \cap B_{\text{new}}, \quad M_t = M_{t-1} \cup \Delta_{\text{FG}}. \quad (2)$$

For a BG correction,

$$\Delta_{\text{BG}} = B_{\text{prev}} \cap \neg B_{\text{new}}, \quad M_t = M_{t-1} \setminus \Delta_{\text{BG}}. \quad (3)$$

Thus an FG step cannot delete from the current mask and a BG step cannot add to it. The externally maintained state both determines the next current-error scribble and supplies the base mask for composition. Cumulative prompts preserve interaction history without an explicit previous-mask input. Figure 1 summarizes the method.

4.4 Experimental V-native curricula

The learned V-native branch was evaluated as an ablation and is not part of the submitted system. It used a randomly initialized, 102,361,217-parameter **ResidualEncoderUNetOrgan** with CT, PET, native binary FG scribble, and native binary BG scribble inputs. Training used whole-mask cross-entropy plus soft Dice, SGD with learning rate 10^{-3} , momentum 0.99, Nesterov acceleration, and weight decay 3×10^{-5} . The expanded curriculum comprised 30,000 zero-prompt segmentation updates (Phase A), 20,000 ground-truth-derived scribble updates (Phase B), and 40,000 current-error updates (Phase C). Phase D continued from 90k using second-generation residuals left by the Phase-C model. Its predefined checkpoint rule selected 130k global updates from 90k, 130k, and 170k candidates.

4.5 Training and test parameters

The submitted system received no new training or fine-tuning; loss, optimizer, augmentation, and training hardware are therefore not applicable. Inference uses the released three-dimensional models, foreground threshold 0.5, sliding-window tile step 0.5, Gaussian weighting, and disabled mirroring/test-time augmentation. Step0 averages the two released-folder logits; correction uses one released folder only.

4.6 Code, model, and data availability

Code availability. The project-specific integration code for the submitted system is publicly available under Apache License 2.0:

<https://github.com/shoheyhey53/autopetv-prompt-delta>.

Model availability. Pretrained source and model downloads are referenced through the upstream autoPET-interactive repository [4] and identified by exact checkpoint hashes; they are not redistributed by our repository. The experimental learned models are not components of the submitted system. **Data availability.** Official challenge data were used under their governing access terms and are not redistributed. The production artifact preserves interaction state through the challenge /output contract; parity was verified over 12 states with zero voxel difference. This engineering detail does not alter the scientific method.

5 Results

5.1 Frozen multistep trajectory

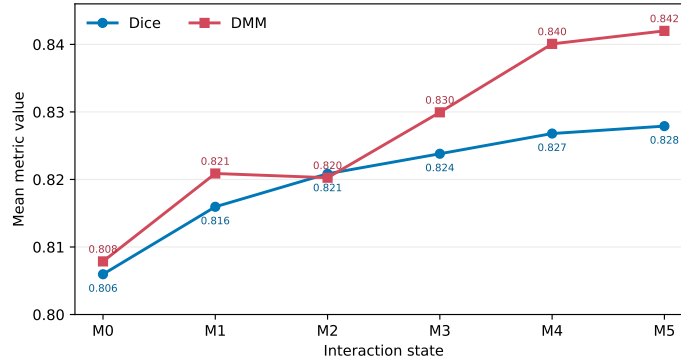
The one-step prompt-delta candidate recovered 49.91% of FG target voxels at 83.54% truth-correction precision and removed 23.76% of BG target voxels at 96.45% precision. Stepwise state retention remained high. The full five-correction trajectory is shown in Table 4 and Fig. 2. Dice increased from 0.805964 to 0.827897, DMM from 0.807865 to 0.842000, FNV fell from 6.1412 to 3.3772 mL, and negative FPV fell from 5.7027 to 5.4118 mL. All-history mean correction retention was at least 99.95% from M_2 onward, with no polarity violations. No-response events nevertheless increased toward later steps, indicating diminishing response to harder residuals rather than state collapse.

5.2 Data scale and V-native learning

The small 48-patient scratch model completed 15,000 stable updates but ended with zero no-prompt Dice, no useful binary FG recovery, and wrong-direction BG behavior. Under the expanded official-data curriculum, Phase A passed its preregistered foundation gate, and the final Phase-C checkpoint reached positive-case median Dice 0.7796. FG probability moved in the correct direction for 36/36

Table 4. Mean frozen prompt-delta trajectory on the 24-case development cohort.

State	Dice	DMM FNV (mL)	Negative FPV (mL)
M_0	0.805964	0.807865	6.1412
M_1	0.815952	0.820883	5.5388
M_2	0.820843	0.820259	4.7302
M_3	0.823805	0.829939	4.3818
M_4	0.826799	0.840077	3.5535
M_5	0.827897	0.842000	3.3772

**Fig. 2.** Dice and DMM across the frozen M_0 – M_5 trajectory.

events and BG for 60/62, with mean shifts of +0.1743 and -0.0670. The learned model therefore acquired lesion segmentation and both prompt polarities.

This success did not imply broad complementarity. Among 18 events with no binary response from released folder 0, eight met the predefined strict-safe rescue criterion with at least one other evaluated released frozen folder, one was uniquely rescued by the V-native model under the same criterion, none were shared, and nine remained unresolved. Phase D increased strict-safe hard-residual rescue from one to three. At the selected 130k checkpoint, FG response occurred in 9/10 events with mean recovery 47.19%, whereas BG response occurred in only 4/31 with mean removal 0.79%; late-BG rescue remained 0/9. Table 5 summarizes the progression.

5.3 Initializer selection

Separate model-specific current-error trajectories compared the final two-folder initializer $I_0 = [0, 5]$ with the canonical all-folder $I_{10} = [0, \dots, 9]$. The smaller initializer was not retained by historical convention: it outperformed the ten-folder alternative on both primary trajectory metrics (Table 6). PSMA AUC-Dice and AUC-DMM changed by -0.285217 and -0.060104 for I_{10} relative to I_0 , respectively.

Table 5. V-native learning progression. Learned models were experimental and not submitted.

Stage	Scale	Updates	Main observation
Small scratch	48 patients	15k	no useful foundation; FG weak and BG reversed
Expanded A	1,477 studies	30k	positive median Dice 0.7086; foundation gate passed
Expanded A-B-C	same	90k	positive median Dice 0.7796; FG 36/36, BG 60/62 directions correct
Phase D selected	same	130k	global three strict-safe hard-residual rescues; late BG still weak

Table 6. Initializer comparison on development trajectories.

Metric	$I_0 = [0, 5]$	$I_{10} = [0, \dots, 9]$	$I_{10} - I_0$
Mean raw AUC-Dice	4.204458	4.087093	-0.117364
Median raw AUC-Dice	4.433874	4.277567	-0.156306
Mean raw AUC-DMM	4.034404	3.938370	-0.096034
Median raw AUC-DMM	4.490909	4.360160	-0.130749

5.4 Complete-trajectory model selection

On the development cohort, frozen folder 0 alone Pareto-dominated the three predefined multi-corrector arms on AUC-Dice and AUC-DMM (Table 7). In particular, adding the selected learned corrector reduced AUC-Dice by approximately 0.16966 and AUC-DMM by 0.19805. The learned model had improved selected local residuals, yet reduced complete-trajectory quality.

Only the single frozen corrector and the simplest distinct challenger (F0+F1) were evaluated in the prespecified comparison on the patient-disjoint confirmation cohort. AUC-Dice was effectively tied, while the additional corrector materially reduced AUC-DMM (Table 8). The slight FPV improvement did not compensate for the lesion-matching deterioration. The challenger-minus-baseline DMM difference was already -0.022589 at M_1 and increased in magnitude to -0.053633 at M_5 , demonstrating accumulation across interactions. Because the challenge gives equal weight to the Dice- and DMM-based ranking components, we selected Baseline A, the single-corrector architecture.

6 Discussion

The central result is not simply that a frozen baseline prevailed. The frozen network contained a strong prompt-conditioned directional response, and state-preserving composition converted that response into a stable state-preserving trajectory without training or an explicit previous-mask channel. Earlier corrections were retained and both Dice and DMM improved over five interactions.

Table 7. Development-cohort complete-trajectory comparison with the frozen initializer.

Corrector arm	Mean AUC-Dice	Mean AUC-DMM	Negative FPV AUC
F0	4.20446	4.03440	221.440
F0 + learned 130k	4.03479	3.83635	220.926
F0 + F1	4.13946	3.77164	221.035
F0 + F1 + learned 130k	4.00011	3.75494	220.564

Table 8. Patient-disjoint confirmation-cohort comparison (24 cases).

Metric	F0 baseline	F0+F1 challenger	Difference
Mean raw AUC-Dice	4.098631	4.099402	+0.000771
Median raw AUC-Dice	4.119477	4.120868	+0.001391
Mean raw AUC-DMM	4.122456	3.945331	-0.177125
Median raw AUC-DMM	4.085426	4.005208	-0.080218
Negative FPV raw AUC	27.8578	27.6743	-0.1835

The remaining failure mode shifted toward absent binary response on increasingly hard residuals.

The learned experiments answer a complementary question. Scaling from 48 patients to more than 1,100 patients enabled a 102M-parameter V-native model to learn both segmentation and prompt semantics. Second-generation self-error training further increased hard-residual response. These are positive learning results. Nevertheless, local rescue counts and target recovery are incomplete selection endpoints. When inserted into the full system, the learned corrector reduced both primary development metrics. Similarly, adding a second pretrained corrector produced small specificity gains but a consistent lesion-level penalty. Additional response diversity can therefore be harmful when responses interact across a stateful trajectory.

DMM was especially informative. On the independent cohort, voxel Dice was essentially unchanged by the second corrector, but DMM deteriorated from the first interaction onward. This suggests that small spatial differences can change lesion detection or matching disproportionately. Because the challenge treats Dice and DMM equally, selection solely by voxel-level overlap, target recovery, or FPV would have chosen the wrong trade-off.

The persistent late-BG limitation also matters. Expanded training learned an average target-direction BG response, but Phase D rarely converted the hardest BG residuals into usable binary deletions. A future model should be designed and selected around complete late-step BG trajectories, rather than by simply increasing model count or training exposure. Our deadline-bounded decision instead favored the simplest configuration supported by both primary metrics.

7 Limitations

The development cohort was repeatedly used for development, and the patient-disjoint confirmation cohort contained only 24 cases. Tracer-by-polarity subgroups were correspondingly small; PSMA-negative state-preservation behavior could not be measured on the confirmation cohort because its negative cases were FDG. Local simulated interactions cannot reproduce every clinician-driven correction, and exact hidden-server AUC semantics were unavailable. The multipoint conversion approximates a scribble with at most ten local EDT sources rather than learning a native line representation. The V-native curricula were single-fold and time-boxed, and no exhaustive fusion search was performed because post-hoc tuning would increase selection bias. Public folder provenance was only partly documented. Finally, no hidden final-test score or rank is claimed.

8 Conclusion

A frozen interactive PET/CT corrector with incremental state updates yielded the most robust complete trajectories in our evaluations. Expanded V-native training and second-generation self-error learning increased local correction responsiveness, but those gains did not consistently improve AUC-Dice and AUC-DMM. Interactive segmentation methods should therefore be selected over complete stateful trajectories, with lesion-level detection matching evaluated alongside voxel overlap.

Acknowledgments. The author thanks the autoPET V organizers and the developers of the public evaluation and pretrained-model resources used in this work.

Competing interests. The author declares no competing interests.

Funding. No external funding was declared for this work.

Data availability. The study used official autoPET challenge data under the applicable access terms. No medical images, labels, patient-level manifests, or identifiers are included in the manuscript source package.

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